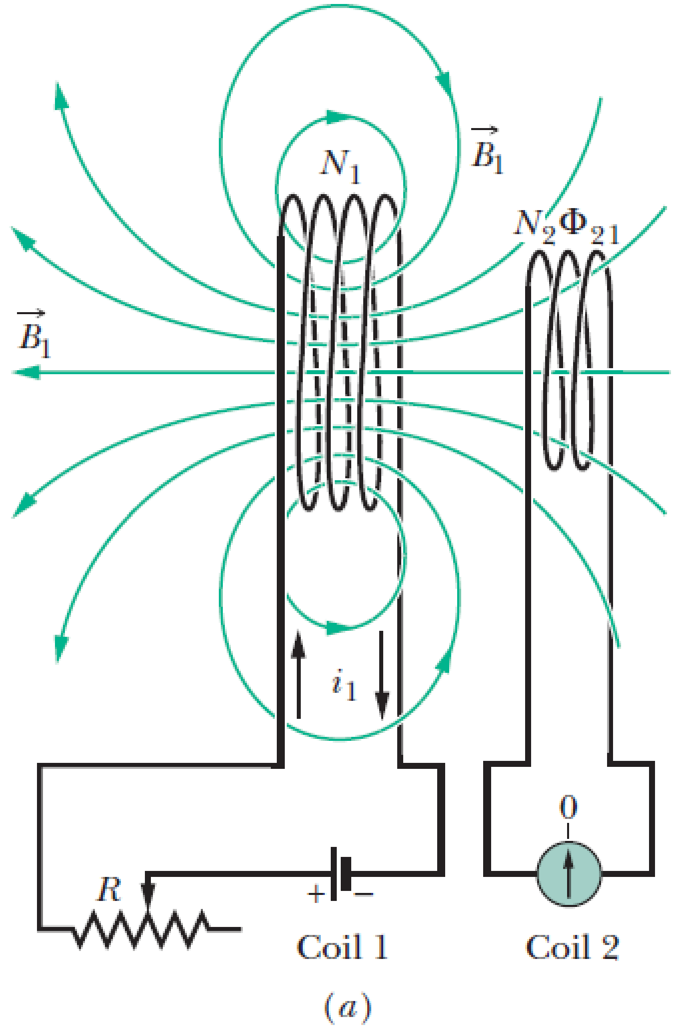
Inductance

Mutual Inductance

Interaction between two loops :

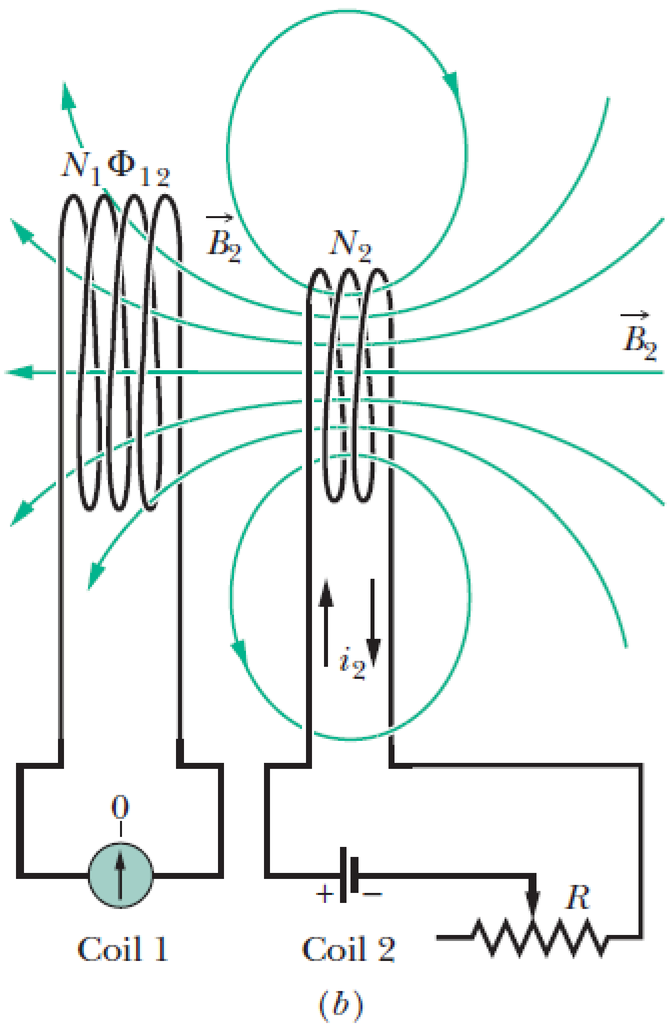
- The magnetic field $\mathbf{B}_1$, produced by current I_1 in coil 1 extends through coil 2.
- If I_1 is varied, *an emf is induced in coil 2* and current registers on the meter connected to coil 2.



Mutual Inductance

Interaction between two loops :

- The roles of the coils interchanged: The magnetic field $\mathbf{B}_2$, produced by current I_2 in coil 2 extends through coil 1.
- If I_2 is varied, *an emf is induced in coil 1*.



Mutual Inductance

Interaction between two loops :

- *The field produced by coil* (1):

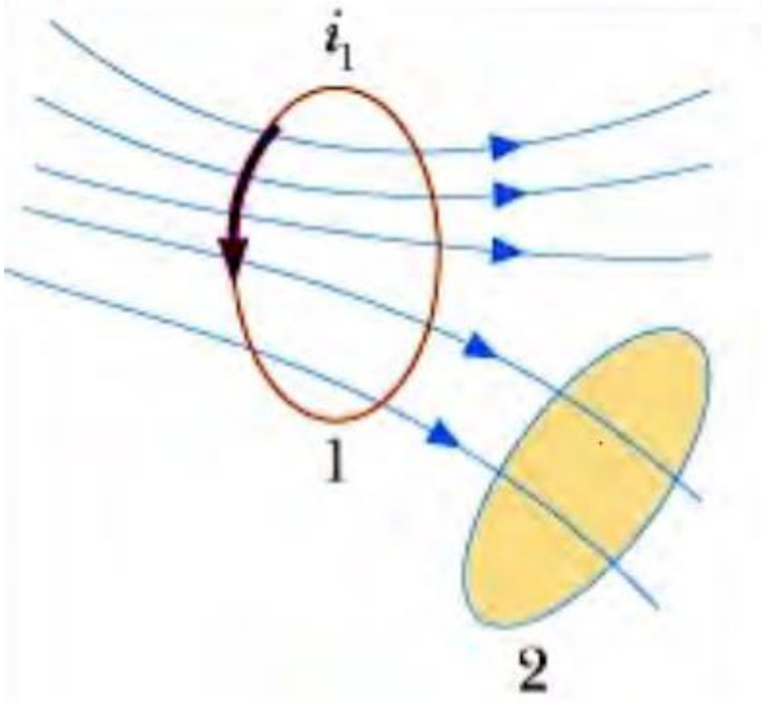
$$\vec{\boldsymbol{B}} = k_m \int \frac{I\, d\vec{\mathbf{L}} \times \vec{\boldsymbol{r}}}{r^3}$$

- *The* ***flux*** *of the field B_1 through the second coil* (2):

$$\Phi_2 = \iint \vec{B}_1 \cdot d\vec{A}_2$$

$$= M_{21}\, I_1$$

$M_{21} =$ Mutual inductance



Mutual Inductance

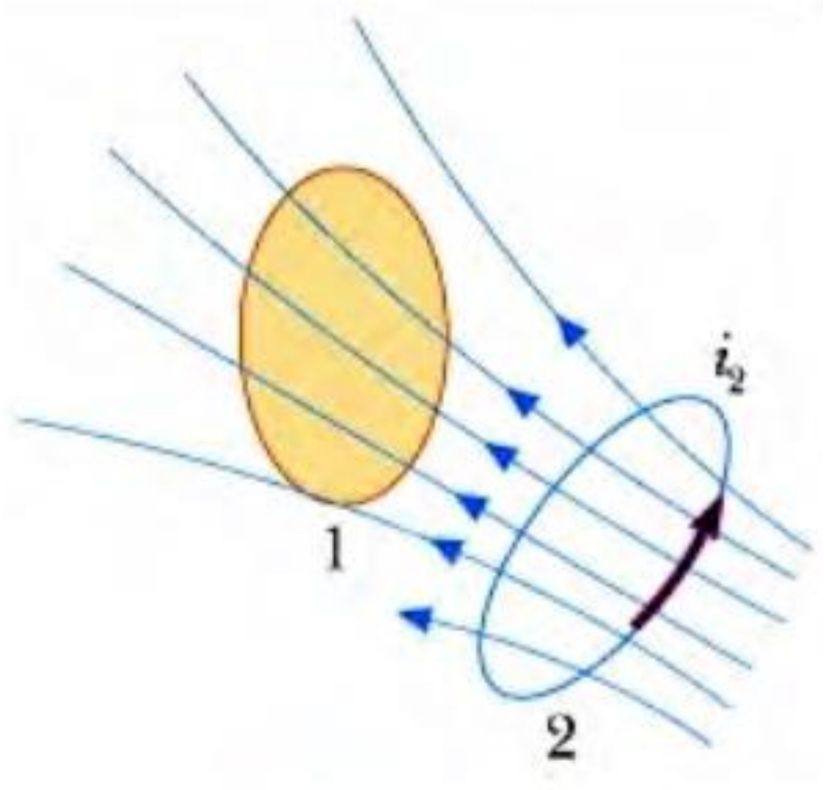
Interaction between two loops :

- *Flux of the field B_2 through the first coil (1) :*

$$\Phi_1 = M_{12}\, I_2$$

- *The quantities* M_{12} , M_{21} *are determined by the geometry of the setup:*

$$M_{21} = M_{12} = M$$



Mutual Inductance

The Mutual Inductance:

- *The total flux through each coil :*

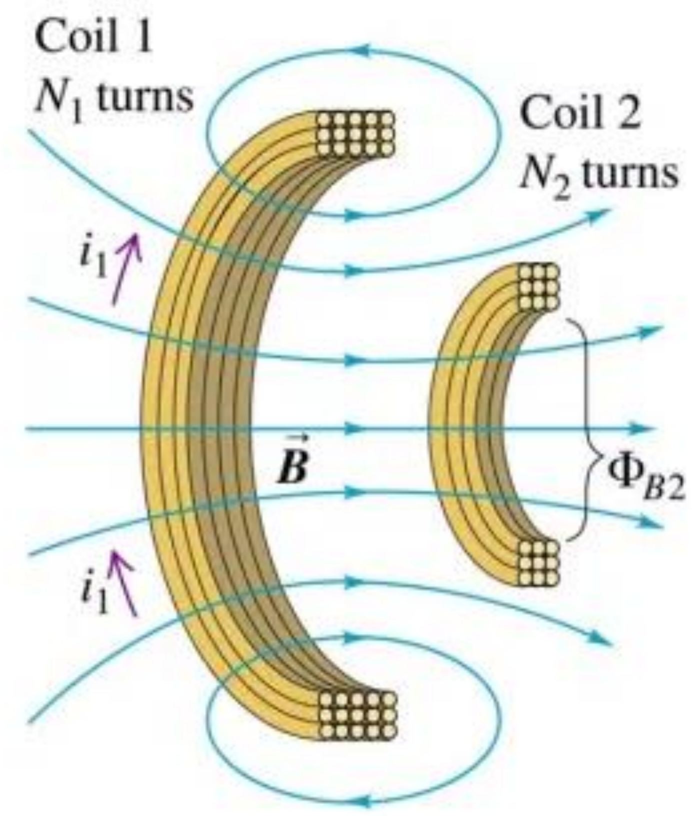
$$N_1\Phi_1 = MI_2$$

$$N_2\Phi_2 = MI_1$$

- *The mutual inductance* M *:*

$$M = \frac{N_1\Phi_1}{I_2} = \frac{N_2\Phi_2}{I_1}$$

Units: Weber/Ampere = Henry



Mutual Inductance

The Mutual Inductance:

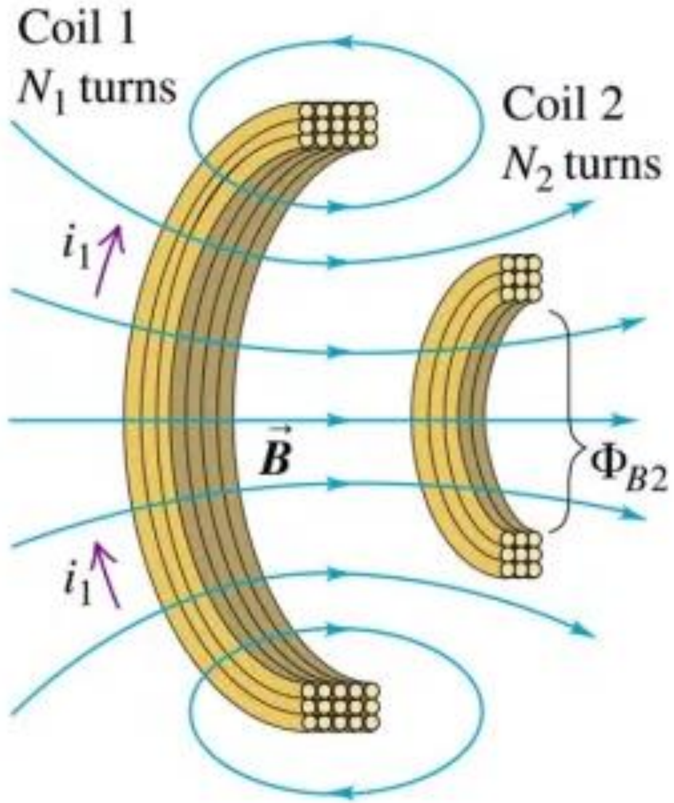
When the current in one coil changes:

- *The induced emf in the other coil :*

$$\mathcal{E}_2 = -N_2 \frac{d\Phi_2}{dt} = -M \frac{dI_1}{dt}$$

- *Likewise:*

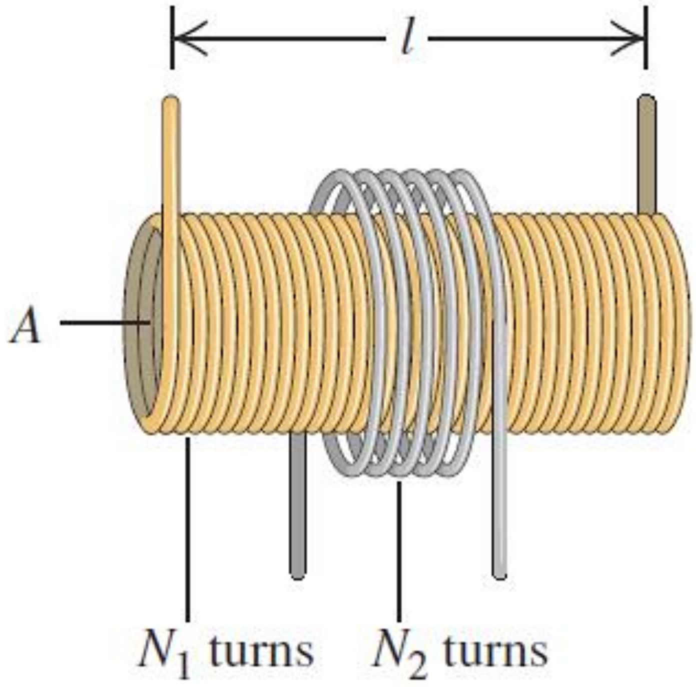
$$\mathcal{E}_1 = -M \frac{dI_2}{dt}$$



Mutual Inductance

Example: Calculating mutual inductance

In one form of a *Tesla coil*, a long solenoid with length ℓ and cross-sectional area A is closely wound with N_1 turns of wire. A coil with N_2 turns surrounds it at its center. Find the mutual inductance *M*.



Mutual Inductance

Example: Calculating mutual inductance

Solution:

$$B_1 = \mu_0 n_1 i_1 = \frac{\mu_0 N_1 i_1}{l}$$

$$M = \frac{N_2 \Phi_{B2}}{i_1} = \frac{N_2 B_1 A}{i_1} = \frac{N_2}{i_1}\frac{\mu_0 N_1 i_1}{l}A = \frac{\mu_0 A N_1 N_2}{l}$$

For: ℓ=0.50 m; A=0.001 m^2; N_1 =1000 turns; N_2 =10 turns

$$M = \frac{(4\pi \times 10^{-7}\ \mathrm{Wb/A \cdot m})(1.0 \times 10^{-3}\ \mathrm{m}^2)(1000)(10)}{0.50\ \mathrm{m}}$$

$$= 25 \times 10^{-6}\ \mathrm{Wb/A} = 25 \times 10^{-6}\ \mathrm{H} = 25\ \mu\mathrm{H}$$

Self-Inductance and Inductors

Self-inductance :

For a single isolated coil:

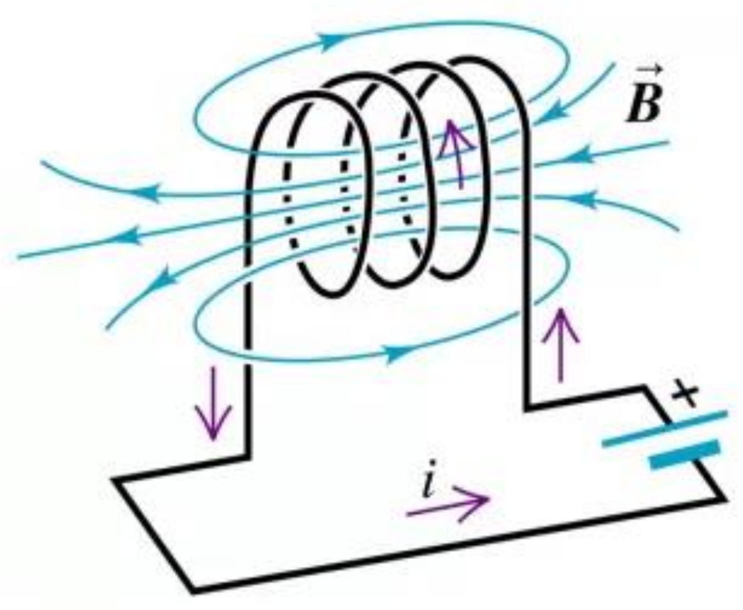
- The coil's *own magnetic field* causes a magnetic flux through the same coil.
- The total "self" flux:

$$N\Phi_m = L\,I$$

- The self-induced emf:

$$\mathcal{E}_1 = -L\,\frac{dI}{dt}$$

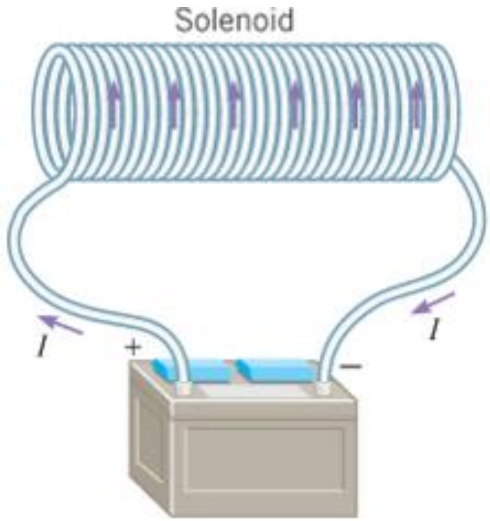
$L =$ Self-inductance



Self-Inductance and Inductors

Example: Ideal solenoid

Self-inductance of a solenoid of radius a with $n = N/\ell$ turns per unit length.



- *Field of solenoid:*

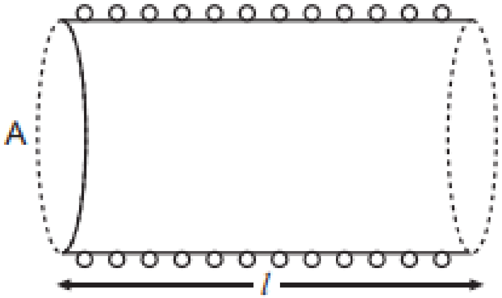
$$B = \mu_0 n\, I$$

- *Flux through one turn:*

$$\Phi_1 = (\mu_0 n\, I)(\pi a^2)$$

- *Self-inductance:*

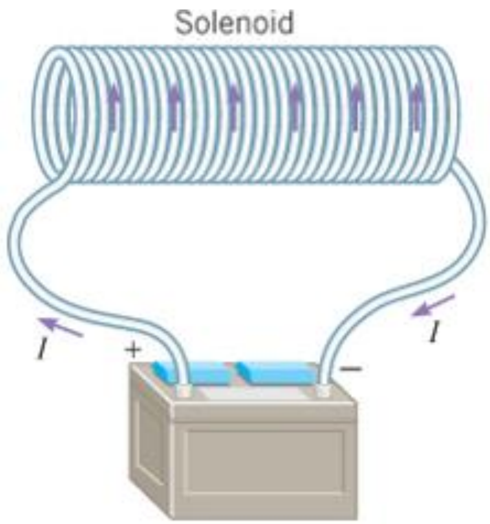
$$L = \frac{N\Phi_1}{I} = \mu_0 n^2 \ell\, (\pi a^2)$$



Self-Inductance and Inductors

Example:

Given: n = 200 *turns/m; A* = 5.0 *cm*^{2}



$$L = (4\pi \times 10^{-7})(200)^2(5.0 \times 10^{-4})\ell$$
$$= (2.5 \times 10^{-5})\ell$$

- *Inductance per unit length:*

$$L/\ell = 2.5 \times 10^{-5}\ \text{H/m}$$

Self-Inductance and Inductors

Inductors as Circuit Element:

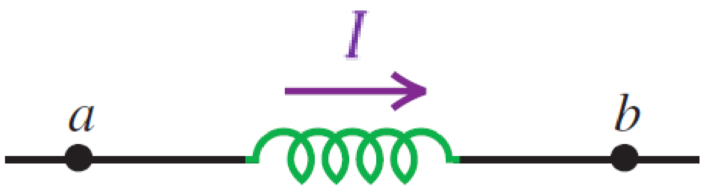
- Self-induced emf (voltage):

$$\mathcal{E}_1 = -L\,\frac{dI}{dt}$$

- Voltage drop:

$$V_{ab} = V_a - V_b$$

$$= L\,\frac{dI}{dt}$$

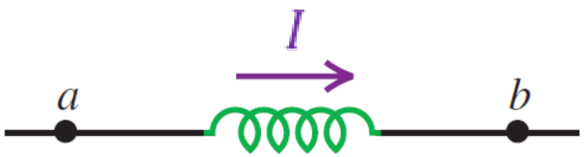


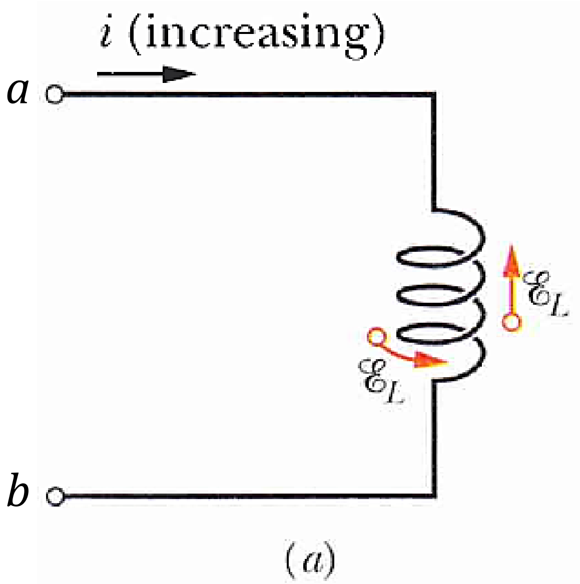
Self-Inductance and Inductors

Inductors as Circuit Element:

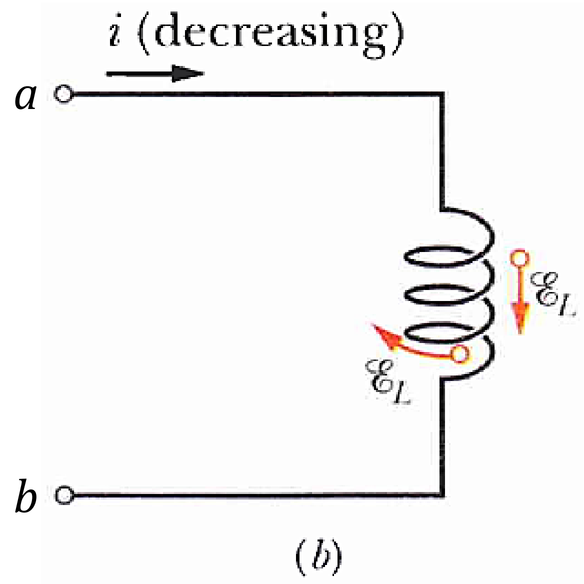
- *When current I changes:*

$$V_{ab} = L\,\frac{dI}{dt}$$





$V_{ab} > 0$



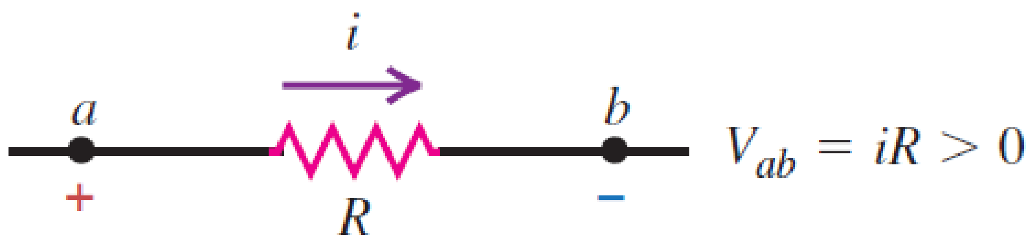
$V_{ab} < 0$

Self-Inductance and Inductors

Inductors as Circuit Element:

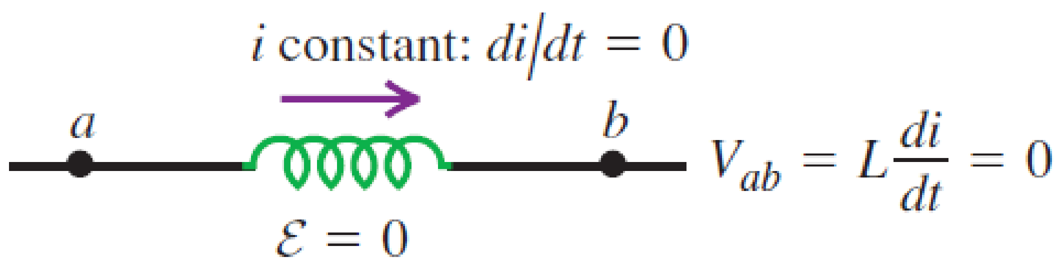
- *Voltage for a resistor :*

(a) Resistor with current i flowing from a to b: potential drops from a to b.



- *Voltage for an inductor:*

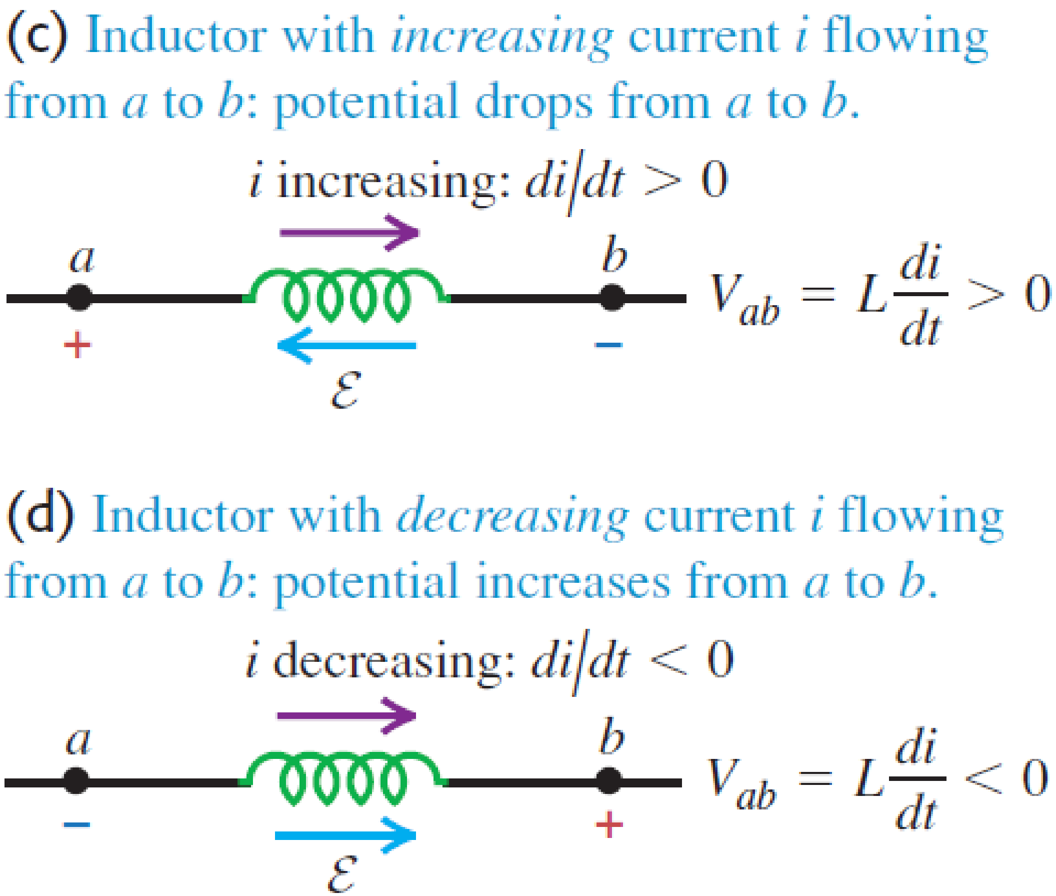
(b) Inductor with *constant* current i flowing from a to b: no potential difference.



Self-Inductance and Inductors

Inductors as Circuit Element:

- *Voltage for an inductor:*



Self-Inductance and Inductors

Example:

If the current in the solenoid of L=40 μH increases uniformly from 0 to 6.0 A in 3.0 μs, find the magnitude and direction of the self-induced emf.

Solution:

$$\frac{dI}{dt} = \frac{6.0\ \mathrm{A}}{3.0 \times 10^{-6}\ \mathrm{s}} = 2.0 \times 10^{6}\ \mathrm{A/s}$$

- *Inductor voltage:*

$$V_{ab} = L\,\frac{dI}{dt}$$

$$= (40 \times 10^{-6}\ \mathrm{H})(2.0 \times 10^{6}\ \mathrm{As}) = 80\ \mathrm{V}$$

Magnetic-Field Energy

Energy Stored in an Inductor

- *Electrical power:*

$$P = \frac{dU}{dt} = IV_{ab} = L\,I\,\frac{dI}{dt}$$

- *Total energy needed to set the current to I_f :*

$$U = \int P dt = L \int_0^{I_f} I\,dI$$

$$= \tfrac{1}{2} L\,{I_f}^2$$

*Energy **stored** in the inductor.*

Magnetic-Field Energy

Example: Ideal solenoid

- *Field and inductance:*

$$B = \mu_0 n\, I \qquad L = \mu_0 n^2 (\ell\, A)$$

- *Energy stored:*

$$U = \tfrac{1}{2} L\, I^2 = \tfrac{1}{2} (\mu_0 n^2 \ell A) I^2$$

$$= \tfrac{1}{2} \frac{1}{\mu_0} ({\mu_0}^2 n^2 I^2)(\ell A)$$

- *Energy density:*

$$u = \frac{U}{(\ell A)} = \frac{B^2}{2\mu_0}$$

*Energy **stored** in the inductor.*

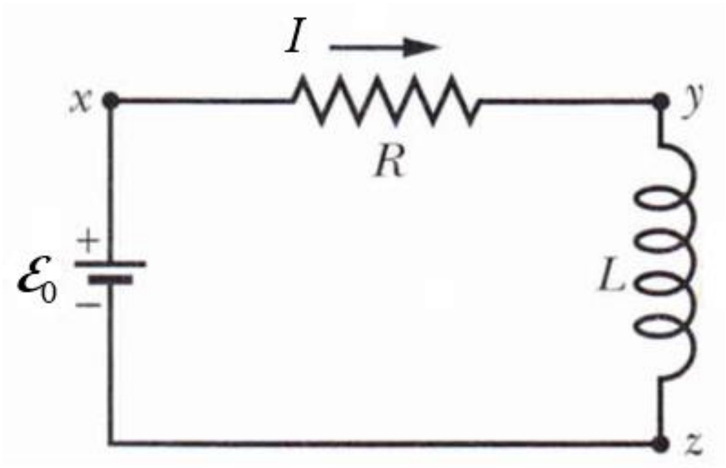
R-L Circuits

Series circuit:

$$\mathcal{E}_0 - V_R - V_L = 0$$

$$\mathcal{E}_0 - IR - L\frac{dI}{dt} = 0$$

$$L\frac{dI}{dt} + IR = \mathcal{E}_0$$



- Solution:

$$I(t) = \frac{\mathcal{E}_0}{R} + ce^{-(R/L)t}$$

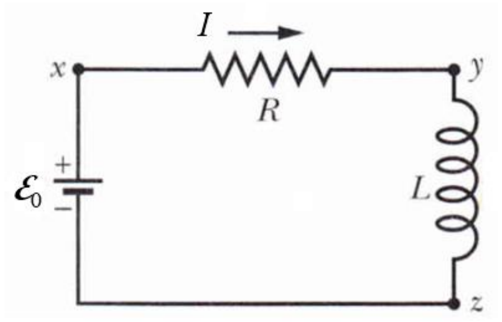
- Given $I = 0$ at $t = 0$:

$$I(t) = \frac{\mathcal{E}_0}{R}\left[1 - e^{-(R/L)t}\right]$$

R-L Circuits

Series circuit:

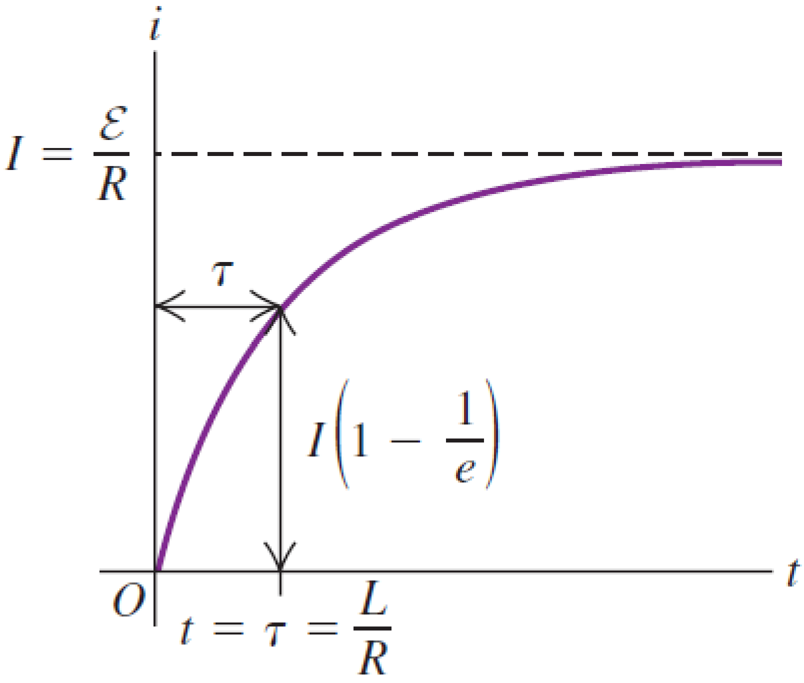
$$I(t) = \frac{\mathcal{E}_0}{R}\left[1 - e^{-(R/L)t}\right]$$



$$I(t) = \frac{\mathcal{E}_0}{R}\left[1 - e^{-(t/\tau)}\right]$$

- The time constant τ:

$$\tau = R/L$$

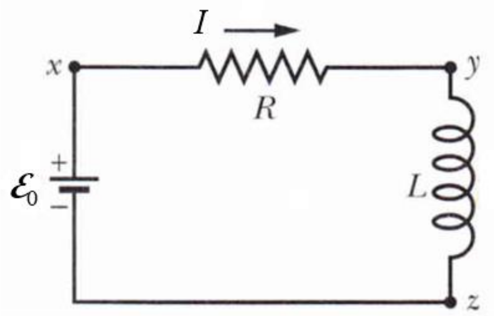


R-L Circuits

Series circuit:

Example:

$$\mathcal{E}_0 = 10\,V\ ;\ R = 2\,k\Omega\ ;\ L = 4.0\,H$$



$$\tau = 2\text{ ms}$$

$$I(t) = (5 \times 10^{-3})[1 - e^{500t}]\text{ A}$$

- The voltages:

$$V_R = IR = 10.0\,[1 - e^{500t}]\text{ V}$$

$$V_L = L(dI/dt) = 10.0\,e^{500t}\text{ V}$$

R-L Circuits

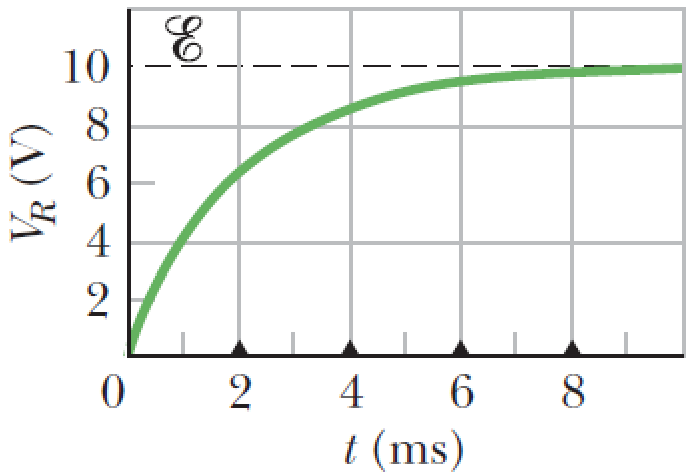
Series circuit:

Example:

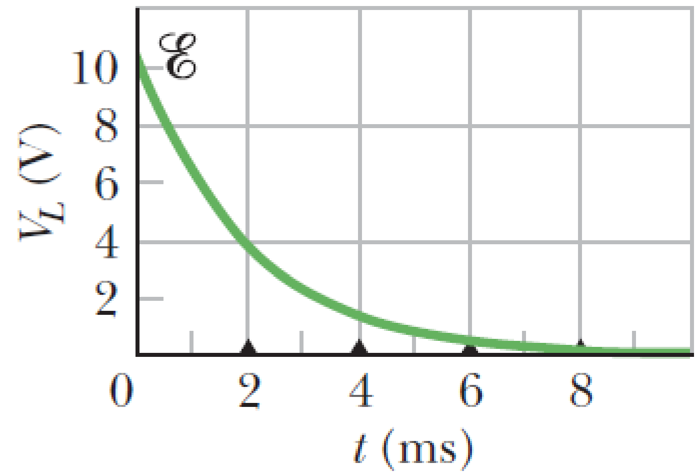
$$\mathcal{E}_0 = 10\,V\ ;\ R = 2\,k\Omega\ ;\ L = 4.0\,H$$

- The voltages:

$$V_R = 10.0\,[1 - e^{500t}]\ \mathrm{V} \qquad V_L = 10.0\,e^{500t}\ \mathrm{V}$$



(a)

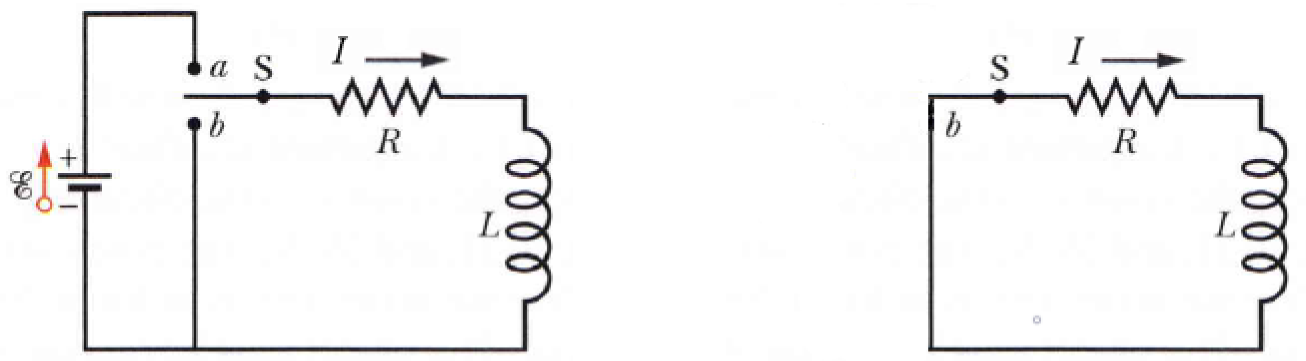


(b)

R-L Circuits

Series circuit – current decay:

Circuit with two-way switch :

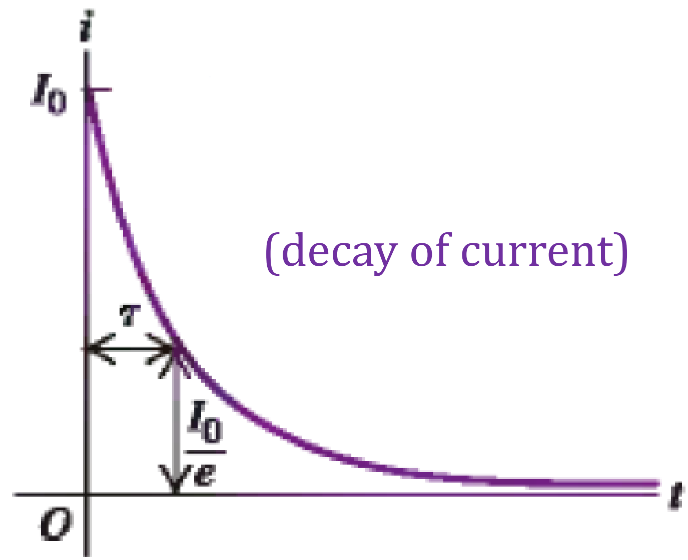


- *Switch at position b:*

$$L\frac{dI}{dt} + IR = 0$$

- *Solution:*

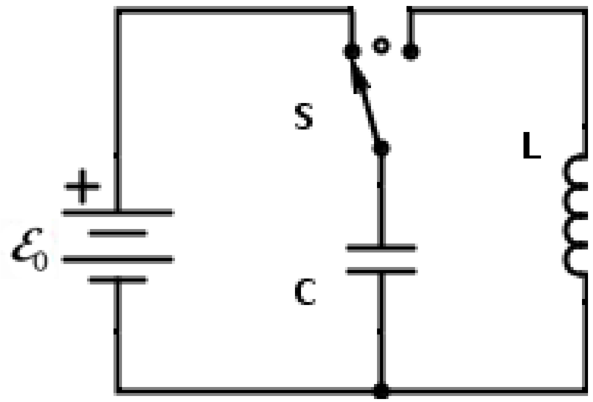
$$I(t) \;= I_0\, e^{-(RL)t}$$



L-C Circuits

***LC Series circuit*:**

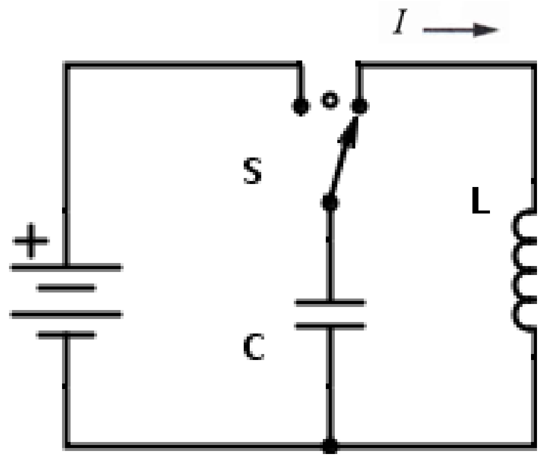
When an initially charged capacitor *C* is connected to an inductor *L.*



- *At t=0 circuit is switched to the right.*

$$V_C = V_L$$

$$\frac{q}{C} = L\frac{dI}{dt}$$



- *But*

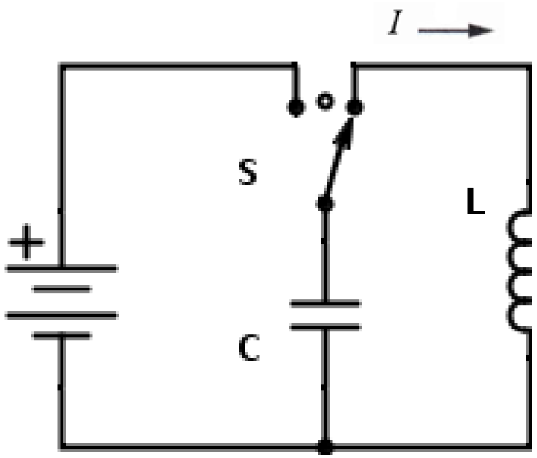
$$I = -\frac{dq}{dt}$$

(*the current discharges the capacitor*)

L-C Circuits

Series circuit:

When an initially charged capacitor *C* is connected to an inductor *L*.



$$\frac{q}{C} = -L\frac{d^2q}{dt^2}$$

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$

Simple harmonic oscillator equation

- *Solution*

$$q(t) = Q\cos(\omega t + \phi) \quad \text{where} \quad \omega = \frac{1}{\sqrt{LC}}$$

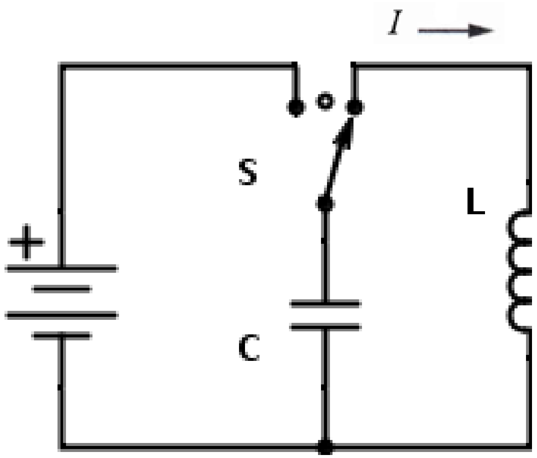
$$(\ Q,\ \phi = \text{constants to be determined}\ )$$

L-C Circuits

Series circuit:

When an initially charged capacitor *C* is connected to an inductor *L.*

$$\frac{d^2q}{dt^2} + \frac{1}{LC}q = 0$$



- *Solution*

$$q(t) = Q\cos(\omega t + \phi)$$

$$I(t) = \omega Q\sin(\omega t + \phi)$$

$$\text{where } \omega = \frac{1}{\sqrt{LC}}$$

L-C Circuits

Example:

A 300-V power supply is used to charge a 25-μF capacitor. After the capacitor is fully charged, it is disconnected from the power supply and connected across a 10-mH inductor. a) Find the angular frequency ω; and b) Find the capacitor charge at t = 1.2 ms.

- *Given*

$$\omega = \frac{1}{\sqrt{LC}} = 2 \times 10^3 \text{ rad/s} \qquad Q_0 = CV_0 = 7.5 \times 10^{-3} \text{ C}$$

$$q(t) = Q\cos(\omega t + \phi) \qquad I(t) = \omega Q\sin(\omega t + \phi)$$

$$(\ f = \omega/2\pi = 320 \text{ Hz} \quad T = 1/f = 3.1 \text{ ms}\ )$$

L-C Circuits

Example:

- *Initial conditions:* At $t = 0$: $Q = Q_0$ $I = 0$

$$q(t) = (0.0075\ \mathrm{C})\cos(2000t) \quad I(t) = (15\ \mathrm{A})\sin(2000t)$$

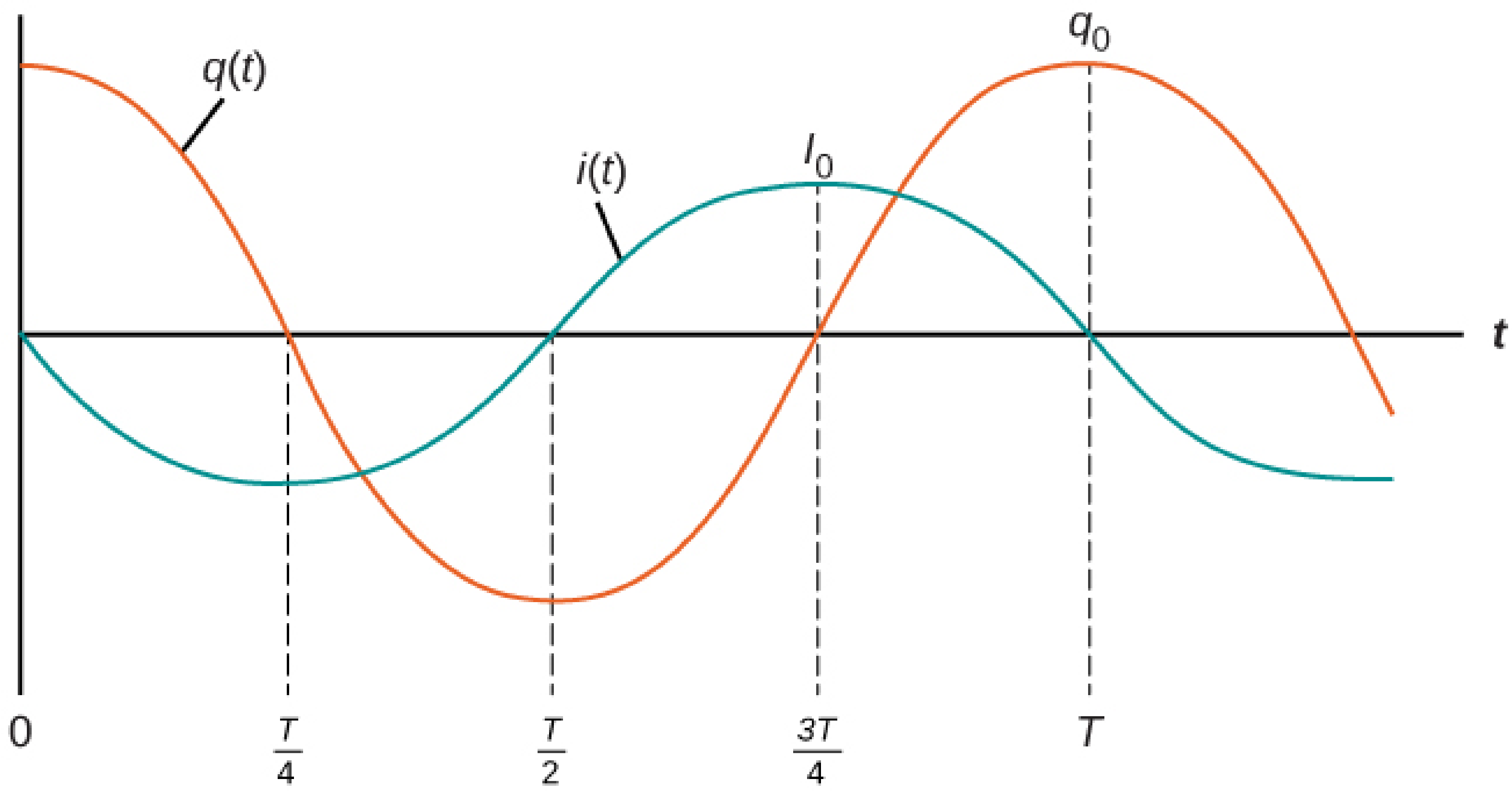
- At $t = 1.2 \times 10^{-3}\ s$

$$q(0.0012\ \mathrm{s}) = -5.5 \times 10^{-3}\ \mathrm{C} \quad I(0.0012\ \mathrm{s}) = 10\ \mathrm{A}$$

L-C Circuits

Series circuit:

$$q(t) = Q_0 \cos(\omega t) \qquad i(t) = -I_0 \sin(\omega t)$$



L-C Circuits

Series circuit: Electrical Energy

$$q(t) = Q_0 \cos(\omega t) \qquad I(t) = I_0 \sin(\omega t)$$

- *Total electrical energy:*

$$U = U_C + U_L = \frac{1}{2C} q^2(t) + \frac{L}{2} I^2(t)$$

$$= \frac{Q_0^2}{2C} \cos^2(\omega t) + \frac{L I_0^2}{2} \sin^2(\omega t)$$

But: $I_0 = \omega Q_0$

$$U = \frac{1}{2}\frac{Q_0^2}{C} = \frac{1}{2} L I_0^2$$

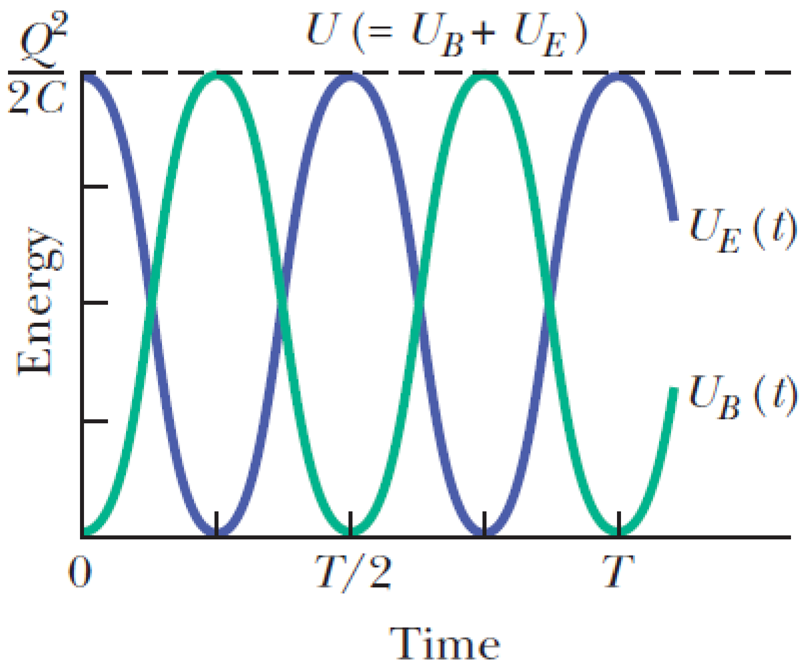
Total energy is constant

L-C Circuits

Series circuit: Electrical Energy

- *Total electrical energy:*

$$U = \frac{1}{2C} q^2(t) + \frac{L}{2} I^2(t)$$



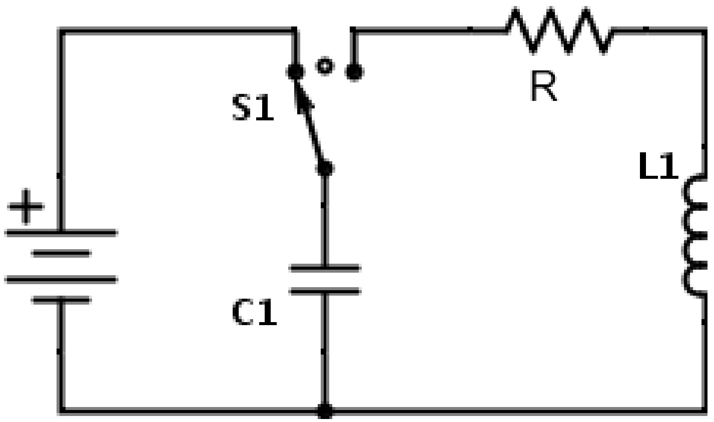
The L-R-C Series Circuit

Charged capacitor C is connected to R and L:

- *At t=0 circuit is switched to the right.*

$$V_C = V_L + V_R$$

$$\frac{q}{C} = L\frac{dI}{dt} + RI$$



- *But* $I = -(dq/dt)$

$$\frac{q}{C} = -L\frac{d^2q}{dt^2} - R\frac{dq}{dt}$$

$$\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{LC}q = 0$$

Damped harmonic oscillator equation

The L-R-C Series Circuit

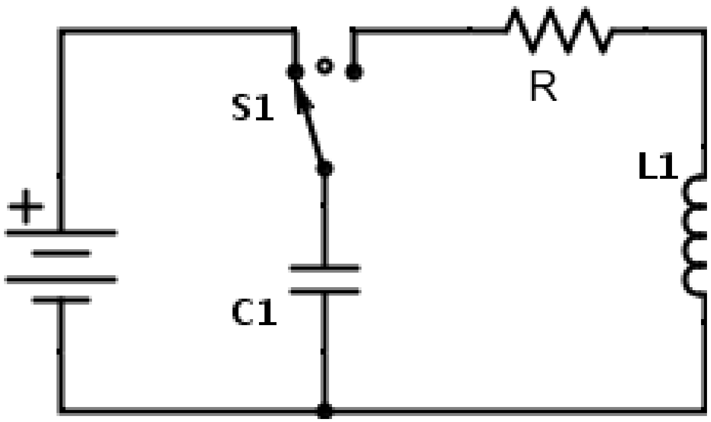
Charged capacitor C is connected to R and L:

$$\frac{d^2q}{dt^2} + \left(\frac{R}{L}\right)\frac{dq}{dt} + \frac{q}{LC} = 0$$

- *Let*

$$\frac{1}{LC} = \omega_0^2 \quad \frac{R}{L} = 2\beta$$

$$\frac{d^2q}{dt^2} + 2\beta\frac{dq}{dt} + \omega_0^2 q = 0$$



- *Solution*

$$q(t) = Ae^{-\beta t}\cos(\omega_1 t + \phi) \quad \text{where} \quad \omega_1 = \sqrt{\omega_0^2 - \beta^2}$$

$$\left(Ae^{-\beta t}\right) = \textit{decaying amplitude}$$

The L-R-C Series Circuit

Damped harmonic oscillation:

$$q(t) = Ae^{-\beta t}\cos(\omega_1 t + \phi)$$

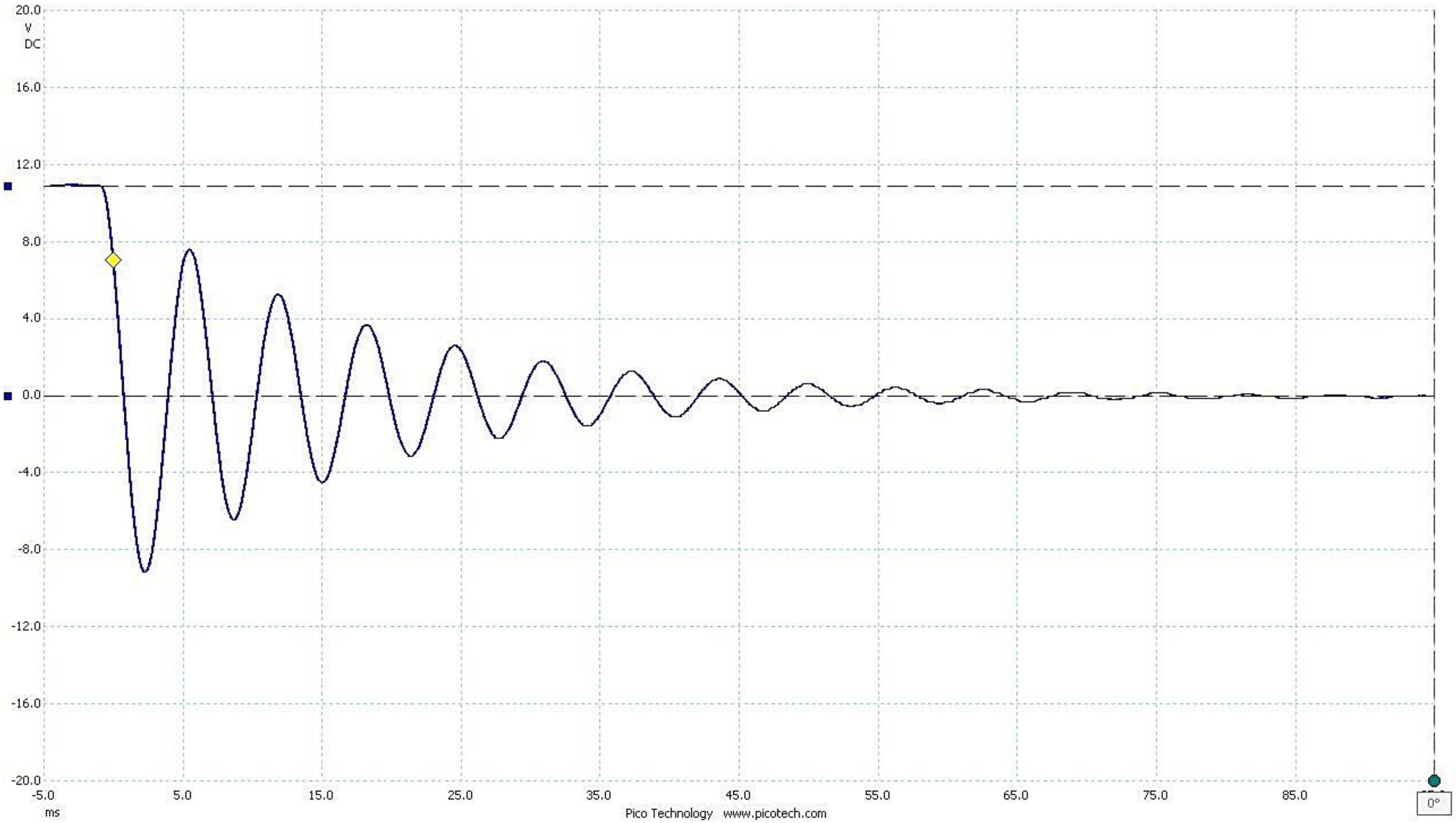


Figure 2. Capacitor: 1007 nF Inductor: 996 mH

The L-R-C Series Circuit

Damped harmonic oscillation:

$$q(t) = Ae^{-\beta t}\cos(\omega_1 t + \phi)$$

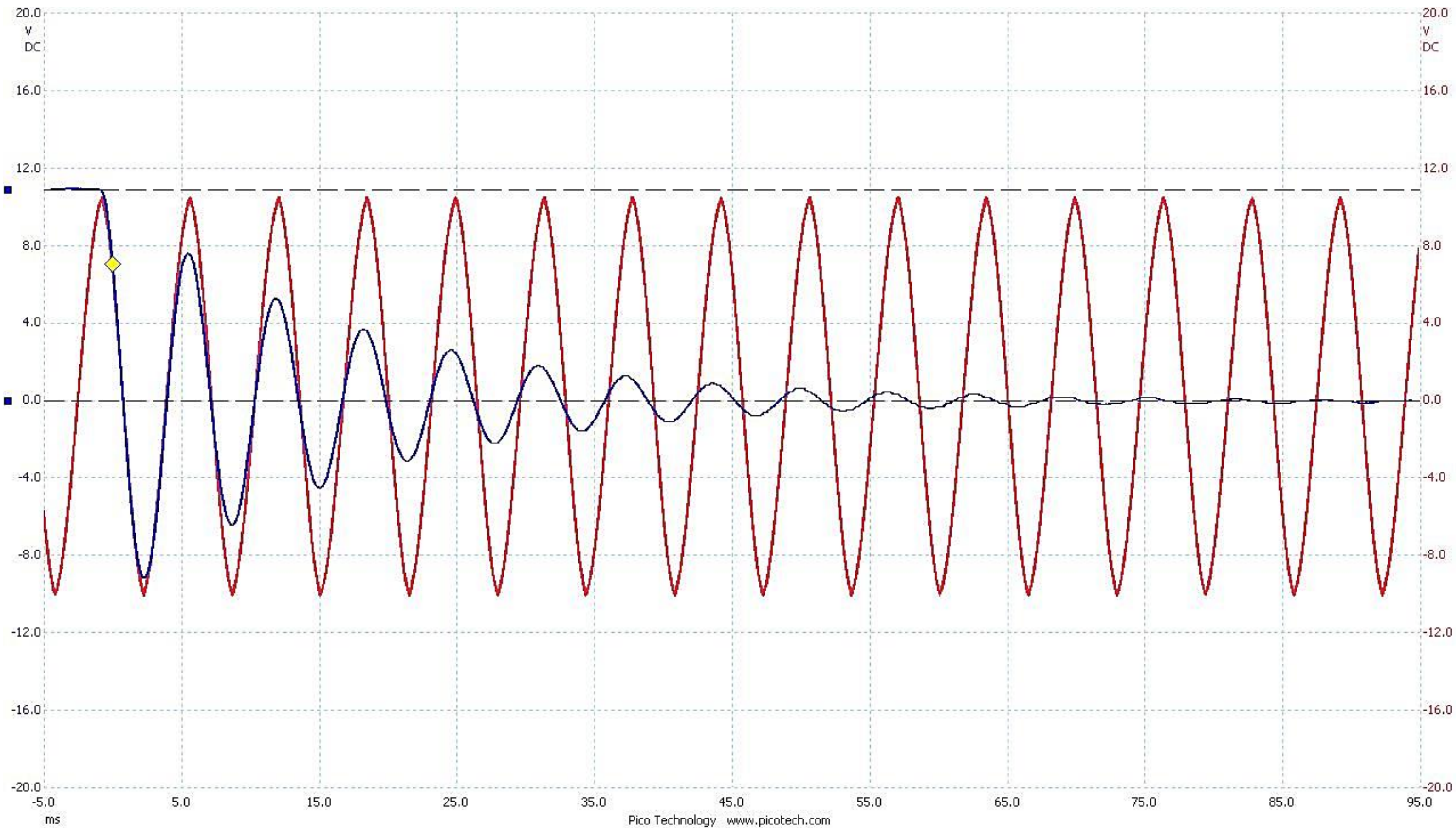


Figure 2. Capacitor: 1007 nF Inductor: 996 mH

The L-R-C Series Circuit

Damped harmonic oscillation:

$$q(t) = Ae^{-\beta t}\cos(\omega_1 t + \phi)$$

- *Energy in the capacitor:*

$$U_C = \frac{1}{2C}q^2(t)$$

$$= \frac{A^2}{2C}e^{-2\beta t}\cos^2(\omega_1 t + \phi)$$

The energy of the electric field oscillates according to a cosine-squared term, and the amplitude decreases exponentially with time.

END